

Angles, polygons, congruence and vectors - Set 1 - Solutions

Specification M5.1-M5.6, M5.10, M5.14, M5.18, M5.19.

Answer key

Question	Answer	Topic
1	F	Angles, polygons, congruence and vectors
2	E	Angles, polygons, congruence and vectors
3	F	Angles, polygons, congruence and vectors
4	D	Angles, polygons, congruence and vectors
5	C	Angles, polygons, congruence and vectors
6	A	Angles, polygons, congruence and vectors
7	E	Angles, polygons, congruence and vectors
8	F	Angles, polygons, congruence and vectors
9	B	Angles, polygons, congruence and vectors
10	C	Angles, polygons, congruence and vectors
11	A	Angles, polygons, congruence and vectors
12	B	Angles, polygons, congruence and vectors

Worked solutions

Question 1 GEO-01 · Angles, polygons, congruence and vectors · spec M5.2

The interior angle of a regular polygon is 156° . How many sides does it have?

Answer: F

Exterior angle 24° ; $360/24 = 15$ sides.

Full method - how and why

Work with the **exterior** angle: they always sum to 360° , which makes the arithmetic trivial.

- 1 Exterior angle $= 180^\circ - 156^\circ = 24^\circ$.

why: Interior and exterior angles at a vertex lie on a straight line.

- 2 The exterior angles of any polygon sum to 360° , and for a regular polygon they are all equal.

why: Walking once around the shape turns you through a full circle.

- 3 Number of sides $= \frac{360}{24} = 15$.

why: Divide the total turn by the turn per vertex.

Where it goes wrong

- Using $\frac{360}{156}$ or $\frac{156}{24}$.
- Using the interior-angle-sum formula and then having to solve $\frac{180(n-2)}{n} = 156$ - correct but slower.

Takeaway For regular polygons, go via the exterior angle: $n = \frac{360}{\text{exterior}}$.

Question 2 GEO-02 · Angles, polygons, congruence and vectors · spec M5.2

A polygon has interior angles that sum to 1980° . How many sides does it have?

Answer: E

$$180(n-2) = 1980 \Rightarrow n-2 = 11 \Rightarrow n = 13.$$

Full method - how and why

Direct application of the interior-angle-sum formula.

- 1 Recall the formula: the interior angles of an n -gon sum to $180(n-2)^\circ$.

why: The polygon splits into $n-2$ triangles, each contributing 180° .

- 2 Set up: $180(n-2) = 1980$.

why: Substitute the given total.

- 3 Divide: $n-2 = 11$, so $n = 13$.

why: $1980 \div 180 = 11$; remember to add the 2 back.

Where it goes wrong

- Forgetting to add 2 and answering 11 - an offered option.
- Using $180n$ instead of $180(n-2)$.

Takeaway Interior sum $= 180(n-2)$: divide, then add 2.

Question 3 GEO-03 · Angles, polygons, congruence and vectors · spec M5.19

$\mathbf{a} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$. Find $2\mathbf{a} - \mathbf{b}$.

Answer: F

$$\begin{pmatrix} 4 \\ -2 \end{pmatrix} - \begin{pmatrix} -3 \\ 4 \end{pmatrix} = \begin{pmatrix} 7 \\ -6 \end{pmatrix}.$$

Full method - how and why

Column-vector arithmetic: operate on each component separately.

1 Double $\mathbf{a}$: $2\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$.

why: A scalar multiplies both components.

2 Subtract $\mathbf{b}$ component by component: $\begin{pmatrix} 4 - (-3) \\ -2 - 4 \end{pmatrix}$.

why: Subtracting a negative adds - the top component becomes $4 + 3$.

3 Simplify: $\begin{pmatrix} 7 \\ -6 \end{pmatrix}$.

why: Both components computed independently.

Where it goes wrong

- Adding $\mathbf{b}$ instead of subtracting, giving $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ - an offered option.
- Getting the sign of the bottom component wrong: $-2 - 4 = -6$, not $+2$.

Takeaway Vectors combine component by component; watch double negatives.

Question 4 GEO-05 · Angles, polygons, congruence and vectors · spec M5.5

Two similar triangles have corresponding sides of length 6 cm and 9 cm. The smaller triangle has area 20 cm^2 . Find the area of the larger triangle.

Answer: D

$$\text{Area scale factor } \left(\frac{9}{6}\right)^2 = \frac{9}{4}: 20 \times \frac{9}{4} = 45.$$

Full method - how and why

Similar shapes: areas scale as the **square** of the length ratio.

1 Length scale factor: $\frac{9}{6} = \frac{3}{2}$.

why: From the smaller to the larger, so the factor is greater than 1.

2 Area scale factor: $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$.

why: Both dimensions scale, so the area scales by the square.

3 Multiply: $20 \times \frac{9}{4} = 45 \text{ cm}^2$.

why: $20 \div 4 = 5$, then $\times 9$.

Where it goes wrong

- Using the length factor directly: $20 \times \frac{3}{2} = 30$ - an offered option.
- Cubing (that would be for volumes), giving 67.5.

Takeaway Area ratio = (length ratio)²; check the direction of the scale factor.

Question 5 GEO-07 · Angles, polygons, congruence and vectors · spec M5.6

The triangle with vertices $(1, 1)$, $(3, 1)$, $(1, 4)$ is enlarged with scale factor -2 , centre the origin. Which of the following is a vertex of the image?

Answer: C

Multiply each coordinate by -2 : $(-2, -2)$, $(-6, -2)$, $(-2, -8)$.

Full method - how and why

Enlargement with a negative scale factor about the origin: multiply every coordinate by that factor.

- 1 With centre the origin, enlargement by k sends (x, y) to (kx, ky) .

why: Position vectors scale directly when the centre is the origin.

- 2 Apply $k = -2$ to each vertex: $(1, 1) \rightarrow (-2, -2)$; $(3, 1) \rightarrow (-6, -2)$; $(1, 4) \rightarrow (-2, -8)$.

why: A negative factor also rotates the shape through 180° - the image is on the opposite side of the origin.

- 3 Check the options: $(-2, -8)$ appears.

why: Match against the computed image vertices.

Where it goes wrong

- Using $+2$ and choosing $(2, 8)$ - an offered option.
- Multiplying only one coordinate.

Takeaway Enlargement about the origin: multiply both coordinates by k ; a negative k flips the shape through the centre.

Question 6 GEO-08 · Angles, polygons, congruence and vectors · spec M5.4

Which of the following sets of information is NOT sufficient to prove that two triangles are congruent?

Answer: A

SSA is not a congruence condition (it is the ambiguous case of the sine rule); the other four are the standard criteria.

Full method - how and why

The four congruence criteria are SSS, SAS, ASA and RHS. Anything else - notably SSA - is not sufficient.

- 1 Recall the valid criteria: SSS, SAS, ASA (or AAS), RHS.

why: These four are the ones that fix a triangle completely.

- 2 Consider SSA: two sides and a **non-included** angle. This is the ambiguous case of the sine rule, where two different triangles can share the same data.

why: The side opposite the known angle can 'swing' to two positions - exactly the situation in TRI-07.

- 3 So SSA is the odd one out.

why: RHS is really a special case of SSA that works, because the right angle removes the ambiguity.

Where it goes wrong

- Thinking RHS is invalid because it looks like SSA - the right angle makes it unambiguous.
- Confusing AAA (similar but not congruent) with ASA.

Takeaway SSS, SAS, ASA, RHS prove congruence; SSA does not, and AAA gives only similarity.

Question 7 GEO-09 · Angles, polygons, congruence and vectors · spec M5.2

In the diagram (not shown), AB is parallel to CD . A transversal makes an angle of $(3x + 10)^\circ$ with AB and the co-interior angle with CD is $(2x - 5)^\circ$. Find x .

Answer: E

Co-interior angles sum to 180° : $5x + 5 = 180 \Rightarrow x = 35$.

Full method - how and why

Co-interior (allied) angles between parallel lines add to 180° .

- 1 Identify the relationship: co-interior angles lie on the same side of the transversal, between the parallel lines.
why: Their names matter: alternate angles are equal, corresponding angles are equal, co-interior angles are supplementary.
- 2 Write the equation: $(3x + 10) + (2x - 5) = 180$.
why: Supplementary means summing to 180° .
- 3 Simplify: $5x + 5 = 180$, so $5x = 175$ and $x = 35$.
why: Combine like terms, then divide.

Where it goes wrong

- Setting the angles equal (treating them as alternate), which gives $x = 15$ - an offered option.
- Arithmetic: $10 - 5 = 5$, not 15.

Takeaway With parallel lines: alternate equal, corresponding equal, co-interior sum to 180° .

Question 8 GEO-10 · Angles, polygons, congruence and vectors · spec M5.3

Which of the following statements is/are true?

- I Every rhombus is a parallelogram.
- II Every rectangle is a rhombus.
- III The diagonals of a kite are perpendicular.

Answer: F

A rhombus has two pairs of parallel sides (I). A rectangle need not have equal sides (II false). The diagonals of a kite cross at right angles (III).

Full method - how and why

Quadrilateral classification. Each claim is about whether one family is contained in another.

- 1 I: a rhombus has two pairs of parallel sides, which is the definition of a parallelogram. **True.**
why: Rhombus = parallelogram with all sides equal, so it is a special parallelogram.
- 2 II: a rectangle has equal **angles** but not necessarily equal sides - a 2×5 rectangle is not a rhombus. **False.**
why: Only the square lies in both families.
- 3 III: a kite's diagonals meet at right angles (the axis of symmetry bisects the other diagonal perpendicularly). **True.**
why: This is a standard kite property.
- 4 I and III only.
why: Match to the option list.

Where it goes wrong

- Reversing II: every rhombus is not a rectangle either, but the claim as stated is about rectangles.
- Thinking a kite's diagonals **bisect** each other - only one bisects the other.

Takeaway Square = rectangle $\cap$ rhombus; kites have perpendicular diagonals but only one is bisected.

Question 9 GEO-12 · Angles, polygons, congruence and vectors · spec M5.6

A shape is reflected in the line $y = x$ and then rotated 90° anticlockwise about the origin. The point $(3, 1)$ ends up at

Answer: B

Reflection in $y = x$: $(3, 1) \rightarrow (1, 3)$. Rotation 90° anticlockwise: $(x, y) \rightarrow (-y, x)$, so $(1, 3) \rightarrow (-3, 1)$.

Full method - how and why

Two transformations in sequence. Apply each rule in the stated order to the coordinates.

- 1 Reflection in $y = x$ swaps the coordinates: $(3, 1) \rightarrow (1, 3)$.
why: The line $y = x$ exchanges the roles of the two axes.
- 2 Rotation 90° anticlockwise about the origin sends $(x, y) \rightarrow (-y, x)$.
why: Worth memorising; check it on $(1, 0) \rightarrow (0, 1) \checkmark$.
- 3 Apply it: $(1, 3) \rightarrow (-3, 1)$.
why: Here $x = 1$, $y = 3$, so the image is $(-3, 1)$.

Where it goes wrong

- Applying the transformations in the wrong order.
- Using the clockwise rule $(x, y) \rightarrow (y, -x)$, giving $(3, -1)$ - an offered option.

Takeaway Anticlockwise 90° : $(x, y) \rightarrow (-y, x)$. Clockwise: $(x, y) \rightarrow (y, -x)$. Order matters.

Question 10 GEO-13 · Angles, polygons, congruence and vectors · spec M5.2

In triangle ABC , angle A is twice angle B , and angle C is 20° more than angle A . Find angle B .

Answer: C

$$B + 2B + (2B + 20) = 180 \Rightarrow 5B = 160 \Rightarrow B = 32^\circ.$$

Full method - how and why

Express all three angles in terms of one, then use the angle sum of a triangle.

- 1 Let angle $B = b$. Then $A = 2b$ and $C = A + 20 = 2b + 20$.
why: Express everything in terms of the angle being asked for.
- 2 Angle sum: $b + 2b + (2b + 20) = 180$.
why: The three angles of a triangle sum to 180° .
- 3 Simplify: $5b + 20 = 180$, so $5b = 160$ and $b = 32^\circ$.
why: Check: $32 + 64 + 84 = 180 \checkmark$.

Where it goes wrong

- Setting $C = 2b + 20$ but forgetting that C is 20 more than A , not than B .
- Arithmetic: $180 - 20 = 160$, not 200.

Takeaway Name one angle, express the rest in terms of it, then use the 180° sum.

Question 11 GEO-14 · Angles, polygons, congruence and vectors · spec M5.5

In triangle ABC , D lies on AB and E lies on AC with DE parallel to BC . $AD = 4$, $DB = 6$ and $DE = 5$. Find BC .

Answer: A

Triangles ADE and ABC are similar with ratio $AD : AB = 4 : 10$. $BC = 5 \times \frac{10}{4} = 12.5$.

Full method - how and why

A line parallel to one side creates a similar triangle. Use the **full** side, not just the part.

- 1 $DE \parallel BC$ makes triangles ADE and ABC similar (equal corresponding angles).

why: Parallel lines give equal corresponding angles, which is the AAA similarity condition.

- 2 Compute the ratio: $AB = AD + DB = 4 + 6 = 10$, so $AD : AB = 4 : 10 = 2 : 5$.

why: The ratio must compare corresponding **whole** sides - AD to AB , not AD to DB .

- 3 Scale up DE : $BC = 5 \times \frac{10}{4} = 12.5$.

why: Multiply by the reciprocal of the ratio to go from small to large.

Where it goes wrong

- Using the ratio $4 : 6$ and answering 7.5 - an offered option.
- Dividing instead of multiplying, giving 2.

Takeaway In similar triangles compare corresponding **full** sides: $\frac{AD}{AB}$, not $\frac{AD}{DB}$.

Question 12 X2-20 · Angles, polygons, congruence and vectors · spec M5.3

$ABCD$ is a quadrilateral. P : 'the diagonals of $ABCD$ bisect each other'. Q : ' $ABCD$ is a rectangle'. Which one of the following is true?

Answer: B

Every rectangle is a parallelogram, and a parallelogram's diagonals bisect each other: P is necessary. A non-rectangular parallelogram has bisecting diagonals without being a rectangle: not sufficient.

Full method - how and why

A quadrilateral hierarchy question. Bisecting diagonals characterise parallelograms, a larger class than rectangles.

- 1 Every rectangle is a parallelogram, and a parallelogram's diagonals bisect each other. So $Q \Rightarrow P$: P is **necessary**.

why: Opposite sides equal and parallel forces the diagonals to share a midpoint.

- 2 Is P sufficient? Take a non-rectangular parallelogram, e.g. a rhombus with angles $60^\circ/120^\circ$: diagonals bisect each other, but it is not a rectangle.

why: Bisecting diagonals say nothing about the angles.

- 3 So P is necessary but not sufficient.

why: For a rectangle you also need the diagonals to be **equal** in length.

Where it goes wrong

- Answering 'necessary and sufficient' by confusing 'bisect each other' with 'bisect each other and are equal'.
- Getting the direction backwards: the broader property (parallelogram) is necessary for the narrower one (rectangle).

Takeaway Diagonals bisect each other $\Leftrightarrow$ parallelogram; add equal diagonals for a rectangle, add perpendicular for a rhombus.